

# Quantitative Research Designs

120 minutes · 6 self-check questions

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*The main observational and experimental designs, what each can and cannot establish, and how to choose the weakest design that still answers your question honestly.*

## Learning objectives

- Describe cross-sectional, case-control, cohort and experimental designs
- Match a design to a research question and justify the choice
- Explain why association does not establish causation and what would
- Identify the principal biases threatening each design
- Explain randomisation, blinding and allocation concealment
- Recognise quasi-experimental designs and when they are the honest option

## Observational designs

- Cross-sectional — exposure and outcome measured at the same moment; fast and cheap; gives prevalence but cannot establish which came first
- Case-control — starts from the outcome and looks backward at exposure; efficient for rare outcomes; vulnerable to recall and selection bias; yields an odds ratio
- Cohort — starts from exposure and follows forward to outcome; establishes temporal sequence; expensive and vulnerable to loss to follow-up; yields relative risk and incidence
- Ecological — units of analysis are groups rather than individuals; cheap but subject to the ecological fallacy
- Case series and case reports — descriptive only, no comparison group, hypothesis-generating at best

*Choosing an observational design*

| Design          | Starts from              | Best for                     | Main weakness             |
|-----------------|--------------------------|------------------------------|---------------------------|
| Cross-sectional | A population at one time | Prevalence, associations     | No temporal sequence      |
| Case-control    | The outcome              | Rare outcomes, long latency  | Recall and selection bias |
| Cohort          | The exposure             | Incidence, temporal sequence | Cost, attrition           |
| Ecological      | Groups                   | Population-level hypotheses  | Ecological fallacy        |

## Experimental designs

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- The investigator assigns the exposure, which is what permits causal inference
- Randomisation distributes both known and unknown confounders across arms — its unique strength
- Allocation concealment stops the recruiter from knowing the next assignment, preventing selection bias at entry
- Blinding stops participants, providers or assessors from knowing the assignment, preventing performance and detection bias
- Randomised controlled trials may be parallel, crossover, factorial or cluster-randomised
- Cluster randomisation assigns groups (schools, clinics, villages) and requires a larger sample because responses within a cluster are correlated
- Analyse by intention to treat: participants are analysed in the arm to which they were assigned, whatever they actually received

### IN PRACTICE

Per-protocol analysis, which counts only those who complied, quietly destroys randomisation and reintroduces confounding. Report it as a secondary analysis if at all.

## Quasi-experimental designs

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- Used when randomisation is impossible, unethical or politically unacceptable
- Non-equivalent control group — an untreated comparison group not formed by randomisation
- Interrupted time series — repeated measures before and after an intervention, strong when many time points exist
- Difference-in-differences — compares change over time between exposed and unexposed groups
- Stepped wedge — all clusters eventually receive the intervention, in a randomised order; ethically attractive for programme rollout
- These designs support weaker causal claims and must state the residual threats explicitly

## Causation and bias

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- Association is necessary but not sufficient for causation
- The Bradford Hill considerations — strength, consistency, temporality, biological gradient, plausibility and others — aid judgement but prove nothing individually
- Temporality is the only genuinely necessary condition: the cause must precede the effect
- Selection bias arises when those included differ systematically from those not included
- Information bias arises from measurement error — including recall bias and interviewer bias

- Confounding arises when a third variable explains the apparent relationship
- Confounding can be handled by design (randomisation, restriction, matching) or by analysis (stratification, regression adjustment)

## Worked examples

### One question, four designs, four different answers

*Does using an insecticide-treated net reduce malaria in under-fives? The design chosen changes what can honestly be claimed.*

1. Cross-sectional: survey households, record net use and current infection. Shows association only — sick children may have been given nets because they were sick.
2. Case-control: take children with malaria and matched controls, ask about past net use. Efficient, but caregivers of sick children recall exposure differently (recall bias).
3. Cohort: enrol net users and non-users, follow for a season. Establishes sequence, but families choosing nets differ from those who do not.
4. Cluster-randomised trial: randomise villages to receive nets. Balances known and unknown confounders — and answers the causal question.
5. The trial is strongest and also the slowest and most expensive; the cross-sectional study is fastest and answers least.

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**The point:** Choose the weakest design that still answers your question honestly, then state plainly what it cannot establish. Overclaiming from a cross-sectional study is the most common criticism in peer review.

### Why per-protocol analysis quietly breaks a trial

*In a trial of a six-month TB treatment regimen, 15% of the intervention arm stopped taking the drug. The analyst proposes excluding them "so we measure the drug's true effect".*

1. Ask who stopped. Typically those with worse side effects, more advanced disease, or less social support.
2. Those same characteristics predict worse outcomes independently of the drug.
3. Removing them leaves a healthier intervention arm compared against an intact control arm.
4. Randomisation guaranteed comparable groups at baseline; this exclusion destroys that guarantee.
5. Intention to treat keeps everyone in the arm they were assigned, and answers the question a clinician actually faces: what happens if I prescribe this?

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**The point:** Intention to treat is the primary analysis. Report per-protocol as a secondary sensitivity analysis if at all, and never as the headline result.

## Key terms

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### **Incidence**

The rate of new cases arising in a population over a defined period.

### **Prevalence**

The proportion of a population having the condition at a point in time.

### **Odds ratio**

The ratio of the odds of exposure in cases to the odds in controls; the standard measure from case-control studies.

### **Allocation concealment**

Preventing those enrolling participants from knowing the upcoming assignment.

### **Intention to treat**

Analysing participants in the arm to which they were randomised, regardless of adherence.

### **Ecological fallacy**

Wrongly inferring something about individuals from a relationship observed between groups.

### **Confounding**

Distortion of an exposure-outcome association by a third variable related to both.

## Activity

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You are asked whether a new community health worker programme reduced under-five mortality. Randomisation is refused by the district. Propose two quasi-experimental designs, state what each can and cannot establish, and name the biases you would have to acknowledge.

## Further reading

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- **Rothman — Epidemiology: An Introduction** — Short, rigorous and readable on design and bias.
- **Hulley et al. — Designing Clinical Research** — Practical chapters on each design with worked examples.
- **Schulz, Altman & Moher (2010) — CONSORT 2010 statement** — What a trial must report, and therefore what it must do.

## Self-check questions

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1. A researcher surveys 500 adults, measuring current salt intake and current blood pressure on the same day. The fundamental limitation is that:

- a. The sample is too small
  - b. Temporal sequence cannot be established, so it is unclear which came first
  - c. An odds ratio cannot be calculated
  - d. Ethical approval is not possible
2. For a rare cancer with a 20-year latency, the most efficient design is:
- a. Prospective cohort
  - b. Case-control
  - c. Randomised controlled trial
  - d. Cross-sectional survey
3. The unique advantage of randomisation over statistical adjustment is that it:
- a. Increases the sample size
  - b. Balances unknown and unmeasured confounders as well as known ones
  - c. Removes the need for a control group
  - d. Guarantees blinding
4. In a trial, participants who stopped taking the drug are excluded from analysis. This:
- a. Is correct practice, known as intention to treat
  - b. Breaks randomisation and can reintroduce confounding
  - c. Improves the precision of the estimate
  - d. Is required by CONSORT
5. Countries with higher average chocolate consumption have more Nobel laureates per capita. Concluding that chocolate improves individual cognition commits:
- a. Recall bias
  - b. The ecological fallacy
  - c. Selection bias
  - d. Attrition bias
6. Which condition is genuinely necessary for a causal claim?
- a. A large sample
  - b. Statistical significance
  - c. Temporality — the cause precedes the effect
  - d. Biological plausibility

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## Answers

1. **b.** This is a cross-sectional design. Exposure and outcome are measured simultaneously, so it can show association but cannot establish that the exposure preceded the outcome.
2. **b.** A cohort study would need an enormous sample followed for decades to accumulate enough cases. Case-control starts from existing cases and looks backward, making rare, slow-developing outcomes tractable.
3. **b.** Regression can only adjust for confounders you measured. Randomisation distributes confounders you never thought of, which no analysis can do.
4. **b.** Excluding non-adherers is per-protocol analysis. Adherence is often related to prognosis, so removing those participants destroys the comparability randomisation created. Intention to treat is the primary analysis.
5. **b.** The data describe countries, not people. Inferring an individual-level relationship from a group-level association is the ecological fallacy — and here national wealth confounds both variables.
6. **c.** Temporality is the one Bradford Hill consideration that cannot be dispensed with. The others strengthen a causal argument but none is individually necessary or sufficient.

